

# SI26\_P12: College Event Certificate Eligibility Board

*AI-Assisted Coding Interview Problem*

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## Problem Statement

Build a compact College Event Certificate Eligibility Board for an organizing team reviewing activity participation. It should look up the points and category of each completed activity, aggregate them per participant, and show who has earned enough points while covering every required activity category.

Use one attractive primary screen or report with the fixed activity table, a small editable participant table, an Evaluate action, category and point progress, ordered eligibility results, summary counts, validation, and sample/reset controls. A participant's completed activities may be edited as tokens, a checklist, or one focused form. You may use in-memory data structures, a spreadsheet or notebook, a browser, desktop or mobile tools, or a CLI that produces a clear visual or tabular report. Evaluate local records only; do not add certificate document generation, event registration, payments, accounts, or scheduling.

Use this fixed activity table:

| Activity ID | Activity             | Category | Points |
|-------------|----------------------|----------|--------|
| A01         | Emerging Tech Talk   | LEARN    | 2      |
| A02         | Soldering Mini Lab   | BUILD    | 3      |
| A03         | Project Pitch Circle | SHARE    | 2      |
| A04         | Open Source Clinic   | BUILD    | 2      |

Load these participant records. Each listed activity ID represents one completed participation:

| Participant ID | Participant | Completed activity IDs |
|----------------|-------------|------------------------|
| C01            | Asha        | A01, A02, A03          |
| C02            | Bilal       | A01, A03, A04          |
| C03            | Chen        | A01, A02, A04          |
| C04            | Divya       | A02, A03, A04          |
| C05            | Eshan       | A01, A03               |

A participant is ELIGIBLE only when the completed activities cover LEARN, BUILD, and SHARE and the point total is at least 6. The built-in totals for C01 through C05 are 7, 6, 7, 7,

and 4. C01 and C02 are eligible. C03 is missing SHARE; C04 is missing LEARN; and C05 is missing BUILD and has fewer than 6 points.

## Contracts

- Trim participant and activity IDs. Participant IDs and participant names must be non-empty, and participant IDs must be unique.
- Every completed activity ID must occur in the fixed activity table. A repeated activity ID for the same participant is invalid rather than earning points twice.
- Report INVALID\_PARTICIPANT, DUPLICATE\_PARTICIPANT\_ID, UNKNOWN\_ACTIVITY, or DUPLICATE\_PARTICIPATION with the participant and offending value. Any input error clears result rows and counts from an earlier evaluation.
- For each valid participant, derive a category set from completed activities and sum the fixed activity points. A completed activity contributes its points exactly once.
- Eligibility requires all three categories and a point total greater than or equal to 6. Evaluate both requirements completely rather than stopping at the first failure.
- For an ineligible participant, list missing categories in the exact order MISSING\_CATEGORY: LEARN, MISSING\_CATEGORY: BUILD, MISSING\_CATEGORY: SHARE, followed by POINTS\_BELOW\_6 when applicable. An eligible participant shows no failure reasons.
- Show eligible participants first and ineligible participants second. Within each status, sort by participant ID ascending. An empty completed-activity list is valid and gives 0 points, no covered categories, and all four failure reasons in the contracted order.
- Reset restores the fixed activities and five valid built-in participant rows, then clears validation, results, and counts until Evaluate is used again.

## Acceptance criteria

- **Required:** Load and evaluate the built-in records in one action; show totals 7, 6, 7, 7, 4, with C01 and C02 eligible and summary counts 2 eligible and 3 ineligible.
- **Required:** Show only MISSING\_CATEGORY: SHARE for C03, only MISSING\_CATEGORY: LEARN for C04, and MISSING\_CATEGORY: BUILD followed by POINTS\_BELOW\_6 for C05.
- **Required:** Add only A04 to C05; show its total as 6, all three categories covered, and new counts 3 eligible and 2 ineligible.
- **Required:** Reset, then clear C01's completed activities; show 0 points and the three missing-category reasons followed by POINTS\_BELOW\_6, with counts 1 eligible and 4 ineligible.
- **Required:** Reset, add a second A01 to C01, report DUPLICATE\_PARTICIPATION naming C01 and A01, and show no stale result rows or counts.
- **Required:** Keep activity definitions, participant inputs, category and point progress, ordered reasons, counts, validation, and sample/reset actions synchronized, and

include focused checks for the built-in oracle, exact point boundary, empty completion list, duplicate participation, and ordering.

- **Optional:** Add a compact three-segment category-progress strip driven by the same derived category set.

Use AI coding assistants. Before implementation, create a short plan with 3–5 ordered steps and useful checkpoints. Be prepared to present that plan, explain any changes you made to it, share relevant prompts, summarize your design, and show test evidence such as tests, screenshots, or output samples.

## How You'll Be Evaluated

- **Planning and Solution Presentation:** Present your 3–5-step implementation plan, explain how the work followed or changed that plan, and demonstrate the working solution with clear explanations
- **AI Prompting Strategy:** Show the prompts you used to translate this problem statement into technical specifications for AI assistants
- **Design Constraints and Technology Choices:** Explain the constraints you provided to AI regarding design patterns, technology stack, and architectural decisions
- **AI-Influenced Decision Making:** Discuss trade-offs, assumptions, and how AI recommendations influenced your choices for components, data structures, and implementation approaches
- **Testing and Validation:** Demonstrate how you tested the application covering both typical usage scenarios and edge cases
- **Live Modification Capability:** Be prepared to implement one small modification, and possibly a second if time permits, using AI assistance; keep your development environment ready for focused changes and verification